MECH 191 — Metallurgy

Semester Metallography Portfolio

EXAMET-5-R-600-HD · 50–500× Reflected Light Microscopy

Fall 2026 · Leeward Community College · Instructor: Kai Santos

How to Use This Portfolio

This portfolio contains 15 modules — one per lecture week of MECH 191. Each module is completed during or immediately after class using the shared class specimen kit.

For each module you will: (1) check out the kit, (2) examine assigned specimens on the EXAMET-5 at the listed magnifications, (3) complete the observation table, (4) produce a sketch or photograph of the key microstructural feature, and (5) answer the short-answer questions — then return the kit before leaving class.

Completed modules are kept together and submitted as a single portfolio at the end of Week 16.

Portfolio Rules

- **Kit handling:** Return ALL specimens before leaving class. Never clean with abrasives. Handle by edges only — fingerprints etch and obscure microstructure.
- **Damage:** If a specimen is damaged, notify the instructor immediately.
- **PPE:** Safety glasses and gloves are required during all lab activities.
- **Missed sessions:** Live demonstrations are not repeated for absences (excused or unexcused).

Student Name: ____

Student ID: ___________________________________

Semester / Year: _________________________   Section: _________________________

Safety & PPE Quiz — Complete Before Module 1

Complete this quiz before beginning Module 1 hands-on lab work. Use the Etchant Safety Data Sheets, PPE Reference Guide, and EXAMET-5 Microscope & Etchant Reference pages to answer if needed — this is a comprehension check, not a memory test. Passing score: 80% (15 of 18 correct) at instructor discretion.

Date completed: ________________________________

Score: _______ / 18	**Result (circle): PASS FAIL**	**Instructor initials: _______**

Section A — Etchant & Chemical Hazard Identification

Circle the letter of the best answer.

Q1. Which etchant's diluent is methanol, corrected from an earlier assumption of ethanol?

a) Marble's Reagent
b) Nital 2%
c) Kalling's No. 2
d) Ferric Chloride

Q2. Which two etchants contain hydrofluoric acid (HF)?

a) Nital and Marble's Reagent
b) Keller's Reagent and Kroll's Reagent
c) Kalling's No. 2 and Ferric Chloride
d) Marble's Reagent and Kalling's No. 2

Q3. What specific first-aid item is called out as the skin treatment for HF exposure from Keller's or Kroll's reagent?

a) Apply ice to the area
b) Apply calcium gluconate gel
c) Apply a baking soda paste
d) No further treatment is needed

Q4. What is the signal word on all six etchant SDS documents?

a) Warning
b) Danger
c) Caution
d) Notice

Q5. Why is latex an unacceptable glove material for Keller's or Kroll's reagent?

a) It dissolves on contact with water
b) It does not reliably block hydrofluoric acid
c) It is too expensive for lab use
d) It is uncomfortable to wear for long periods

Q6. Which etchant's SDS lists flammable liquid, skin corrosion, AND toxic-if-inhaled hazard classes together?

a) Marble's Reagent
b) Ferric Chloride
c) Kalling's No. 2 Reagent
d) Keller's Reagent

Q7. Where must an HF-containing etchant be mixed and used?

a) At any open bench
b) In a fume hood
c) Outdoors only
d) In the specimen storage cabinet

Q8. What does the SDS list as a hazard specific to methanol exposure that is not shared by the other diluents on this page?

a) Skin staining
b) Optic nerve damage / risk of blindness
c) Permanent hearing loss
d) Loss of sense of smell

Section B — Match Each Etchant to the Specimen Material It's Used On

Write the letter of the correct specimen material in the "Your Answer" column. Each letter is used exactly once.

Etchant	Your Answer	Specimen Material Options
1. Nital 2%		A. Titanium (Ti-6Al-4V) B. Carbon steel (1018, 1045, 1075) C. Precipitation-hardening stainless (17-4PH) D. Aluminum (6061, 6063, 5052, 2024) E. Brass and copper F. Austenitic stainless steel (304, 316)
2. Marble's Reagent		
3. Kalling's No. 2		
4. Keller's Reagent		
5. Ferric Chloride		
6. Kroll's Reagent		

Section C — PPE Selection

Circle the letter of the best answer.

Q9. What is the minimum eye protection required at all times in the shop, even when only observing?

a) No eye protection needed if not touching anything
b) ANSI Z87.1 safety glasses
c) Reading glasses
d) Sunglasses

Q10. What additional eye/face protection is required, beyond goggles, when using Kroll's reagent?

a) Nothing further — goggles alone are sufficient
b) A full face shield
c) Sunglasses worn over goggles
d) A welding helmet

Q11. What is the correct PPE rule when operating a lathe, mill, or drill press?

a) Wear thick leather gloves for a better grip
b) Wear no gloves at all
c) Wear latex exam gloves
d) Gloves are optional, student's choice

Q12. What glove material is correct for mixing or handling any of the six lab etchants?

a) Latex

b) Cotton
c) Nitrile or neoprene
d) No gloves are required

Q13. What must be staged at the etching station any time Keller's or Kroll's reagent is in use?

a) A fire extinguisher only
b) Calcium gluconate gel
c) An eyewash bottle only, nothing else
d) A spare pair of safety glasses

Q14. Per the welding lens shade chart, what shade range is recommended for SMAW (stick) welding above 160 amperes?

a) 4–6
b) 7–10
c) 10–12
d) 12–14

Section D — Short Answer / Applied Scenarios

Answer in complete sentences on the lines provided.

Q15. A classmate is about to etch a titanium specimen using Kroll's reagent while wearing latex exam gloves and basic safety glasses with no face shield. Identify TWO PPE errors in this scenario and explain the risk each one creates.

Q16. Explain why gloves are removed rather than worn when operating a lathe or drill press, even though gloves protect hands in most other shop contexts.

Q17. Name the specific hazard first-aid item that must be on hand any time Keller's or Kroll's reagent is in use, and explain what it treats and why.

Q18. The original portfolio PPE rule states only "safety glasses and gloves are required." Explain why this is not sufficient for every etching task in this lab, using at least one specific etchant hazard as your example.

Grain boundaries are the 2-D interfaces between adjacent crystals. Their size, shape, and misorientation angle directly control strength, toughness, and ductility — the Hall-Petch relationship states that smaller grains produce stronger metals.

Specimens:

- A – 1018 Low-carbon steel (normalized, etched)

Magnifications:

- 50× – overview scan
- 100× – grain boundary network
- 100× – grain orientation contrast
- 500× – inclusions (MnS stringers)

Etchant: Nital 2% (2% HNO_3 in ethanol)

Equipment & Supplies

- EXAMET-5-R-600-HD metallurgical microscope
- Pre-mounted Specimen A (1018 steel, phenolic mount)
- Lint-free lens tissue
- Cotton gloves or finger cots
- Ruler/scale for field-width grain estimates
- Pencil and this portfolio handout

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – overview scan → 100× – grain boundary network → 100× – grain orientation contrast → 500× – inclusions (MnS stringers)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Grain boundary network visible (50×)
- ☐ Equiaxed grain shape confirmed (100×)
- ☐ Grain orientation contrast (light/dark grains) (100×)
- ☐ MnS inclusion stringers (gray, elongated) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Sketch or describe the grain boundary network at 100×. Are the grains roughly equiaxed or elongated?

Answer:

Q2. Atomic bonding in steel is metallic. What property of metallic bonding allows grain boundaries to exist without the material cracking?

Answer:

Q3. Estimate the average grain diameter at 100× (field width ≈ 2.3 mm on EXAMET-5). Count grains across the field and divide the width by that count.

Answer:

Q4. If you increased to 500×, what new detail appears that was invisible at 100×?

Answer:

Step 4 — Engineering Equations & Concept Check

Hall-Petch Relationship

$$\sigma y = \sigma 0 + ky / \sqrt{d}$$

- σy — yield strength
- $\sigma 0$ — friction stress (intrinsic lattice resistance to dislocation motion)
- ky — Hall-Petch (strengthening) coefficient, material-specific
- d — average grain diameter

Critical-Thinking Questions

C1. Two identical steel samples have the same composition and heat treatment, but Sample X has a finer grain structure than Sample Y. Using the Hall-Petch relationship, explain which sample you expect to have the higher yield strength and why, in terms of dislocation motion.

Answer:

C2. The Hall-Petch relationship predicts that yield strength increases without limit as grain size shrinks toward zero. Is this physically realistic? What do you think eventually limits how much refining the grain size can strengthen a metal?

Answer:

Mechanical properties emerge from microstructure. Pearlite is harder and stronger than ferrite; a finer grain size raises yield strength (Hall-Petch). This module connects what you measure in a test lab to what you see at the microscopic scale.

Specimens:

- A – 1018 steel (normalized)
- B – 1045 steel (normalized), Vickers hardness indent applied for Q2

Magnifications:

- 100× – indent identification and measurement
- 100× – ferrite/pearlite ratio comparison
- 500× – lamellar pearlite spacing

Etchant: Nital 2%

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens A, B (phenolic mounts, Vickers indent applied to B for Q2)
- Stage micrometer or calibrated eyepiece graticule
- Calculator (for HV formula)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – indent identification and measurement → 100× – ferrite/ pearlite ratio comparison → 500× – lamellar pearlite spacing

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ White ferrite grains visible (A vs B comparison) (100×)
- ☐ Dark pearlite colonies visible (100×)
- ☐ Square Vickers indent visible on Specimen B (100×)
- ☐ Both diagonals of indent measurable (100×)
- ☐ Lamellar pearlite lamellae resolvable (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Compare Specimens A and B at 100×. Which has more pearlite? Explain why, given 1045 has more carbon than 1018.

Answer:

Q2. Measure the Vickers indent diagonal on Specimen B (after the indent is applied) at 100×. The field width at 100× is ~2.3 mm. Estimate the diagonal in mm, then calculate: $HV = 1.854 \times F / d_{avg}^2$. (Use F = 1 kgf = 9.81 N; d in mm.)

Answer:

Q3. Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$. If grain size is refined by 50%, what happens to yield strength? (Qualitative answer is fine.)

Answer:

Q4. Which specimen would you expect to have higher tensile strength — and why? Does the microstructure support that prediction?

Answer:

Step 4 — Engineering Equations & Concept Check

Engineering Stress, Strain & Hooke's Law

$$\sigma = F / A0 \qquad \varepsilon = \Delta L / L0 \qquad \sigma = E \cdot \varepsilon \text{ (elastic region only)}$$

- σ — engineering stress; F — applied axial load; $A0$ — original cross-sectional area
- ε — engineering strain; ΔL — change in gauge length; $L0$ — original gauge length
- E — elastic (Young's) modulus

Critical-Thinking Questions

C1. Two tensile specimens are pulled to the same engineering stress — one is pearlitic, one is ferritic. Which one do you expect to show more strain (elongation) at that stress level, and why does the microstructure — not just the stress — determine the answer?

Answer:

C2. Hooke's law ($\sigma = E\varepsilon$) only holds in the elastic region. In words, what physically happens inside the metal's crystal structure once loading passes the yield point that makes stress and strain no longer proportional?

Answer:

The iron-carbon phase diagram predicts the equilibrium ferrite and pearlite fractions at a given carbon content. Comparing three normalized steels spanning the practical carbon range — low-carbon 1018, medium-carbon 1045, and high-carbon 1075 — shows how composition alone controls the ferrite/pearlite balance, and therefore hardness and strength, without any heat treatment involved.

Specimens:

- A – 1018 steel (normalized)
- B – 1045 steel (normalized)
- C – 1075 steel (normalized)

Magnifications:

- 100× – phase field overview
- 100× – ferrite vs. pearlite distinction
- 500× – pearlite lamellar spacing across all three

Etchant: Nital 2%

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens A, B, C (phenolic mounts)
- Iron-carbon phase diagram reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – phase field overview → 100× – ferrite vs. pearlite distinction → 500× – pearlite lamellar spacing across all three

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Predominantly ferrite + minor pearlite (1018) (100×)
- ☐ Roughly equal ferrite/pearlite (1045) (100×)
- ☐ Dominant pearlite with thin ferrite network (1075) (100×)
- ☐ Pearlite lamellae resolvable across all three specimens (500×)

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______×   Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Draw a simple schematic of the iron-carbon phase diagram and mark where 1018, 1045, and 1075 each fall at room temperature after normalizing.

Answer:

Q2. At 100×, rank Specimens A, B, and C by visually estimated pearlite fraction. Does the ranking match what you'd predict from each grade's carbon content using the lever rule?

Answer:

Q3. At 500×, compare pearlite lamellar spacing across the three specimens. Does spacing change meaningfully with carbon content, or is it controlled primarily by cooling rate instead?

Answer:

Q4. 1075 has roughly four times the carbon content of 1018. Would you expect 1075 to have four times the hardness? Explain using what you know about how pearlite fraction relates to hardness.

Answer:

Step 4 — Engineering Equations & Concept Check

Lever Rule — Ferrite/Pearlite Fraction (hypoeutectoid steel)

$$\text{Fraction pearlite} \approx (C0 - 0.022) / (0.76 - 0.022) \qquad \text{Fraction ferrite} = 1 - \text{Fraction pearlite}$$

- C0 — carbon content of the steel, wt%
- 0.022 — maximum carbon solubility in ferrite at room temperature, wt%
- 0.76 — eutectoid composition, wt% C

Critical-Thinking Questions

C1. Using the lever-rule equation, does increasing carbon content from 1018 (~0.18% C) to 1045 (~0.45% C) increase or decrease the pearlite fraction? Does this match what you observed under the microscope?

Answer:

C2. If a steel's carbon content exceeded 0.76%, the equation above would give a pearlite fraction greater than 100%. What actually happens microstructurally past the eutectoid composition that this equation doesn't capture?

Answer:

Stainless steel resists corrosion through a passive chromium oxide layer, but different stainless families reach very different microstructures from that same starting point. 304 and 316 are both austenitic (FCC, non-magnetic) — 316 differs mainly by a molybdenum addition for chloride resistance. 17-4PH is a precipitation-hardening martensitic grade with a completely different strengthening mechanism and microstructure. Comparing all three in one sitting shows that 'stainless' describes a corrosion behavior, not a single microstructure.

Specimens:

- D – 304 austenitic stainless steel
- E – 316 austenitic stainless steel
- F – 17-4PH precipitation-hardening stainless steel

Magnifications:

- 100× – austenite grain structure and twin bands (D, E)
- 100× – martensitic matrix and precipitate distribution (F)
- 500× – twin boundary detail / fine precipitate detail

Etchant: Marble's reagent (D, E): 10g $CuSO_4$ + 50 mL HCl + 50 mL H_2O | Kalling's No. 2 (F): 5g $CuCl_2$ + 100 mL HCl + 100 mL ethanol

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens D, E, F (phenolic mounts)
- PREN / stainless family reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – austenite grain structure and twin bands (D, E) → 100× – martensitic matrix and precipitate distribution (F) → 500× – twin boundary detail / fine precipitate detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Equiaxed austenite grains, 304 (Specimen D) (100×)
- ☐ Equiaxed austenite grains, 316 (Specimen E) (100×)
- ☐ Straight annealing twin bands crossing multiple grains (D, E) (100×)
- ☐ Martensitic matrix, no twinning (17-4PH, Specimen F) (100×)
- ☐ Fine precipitate distribution (Specimen F) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Identify annealing twins in Specimens D and E. Why do FCC austenitic stainless steels twin readily during annealing while the martensitic Specimen F does not?

Answer:

Q2. 316 adds roughly 2–3% molybdenum compared to 304. That addition barely changes what you can see under the microscope, yet 316 is specified over 304 in marine and chloride-heavy environments. What does Mo actually do, and why doesn't its benefit show up as a visible microstructural difference?

Answer:

Q3. Specimen F (17-4PH) looks nothing like D or E under the scope. Describe what you observe instead, and explain why 'stainless' doesn't guarantee a particular microstructure the way '1018' or '1045' does for carbon steel.

Answer:

Q4. 17-4PH is strengthened by precipitation during aging, not by the FCC structure that produces twinning in D and E. Based on what you observed, would you expect 17-4PH to be more or less prone to work hardening during machining than 304? Explain your reasoning.

Answer:

Step 4 — Engineering Equations & Concept Check

Pitting Resistance Equivalent Number (PREN)

$$PREN = \%Cr + 3.3(\%Mo) + 16(\%N)$$

- %Cr, %Mo, %N — weight percent chromium, molybdenum, and nitrogen in the alloy

Critical-Thinking Questions

C1. 316 stainless adds roughly 2–3% Mo compared to 304. Using the PREN formula, estimate how many PREN points that Mo addition is worth, and explain why the formula weights Mo so much more heavily than Cr.

Answer:

C2. 17-4PH is a precipitation-hardening martensitic stainless, not an austenitic grade like 304/316. Is PREN still a meaningful comparison across all three families, or does 17-4PH's different base microstructure change what "corrosion resistance" depends on?

Answer:

Nonferrous Metals & Phase Diagrams

Nonferrous alloys rely on different hardening mechanisms than steel. Aluminum uses precipitation hardening; brass relies on solid-solution strengthening. Both show grain structures at similar magnifications to steel but with very different twin densities — a direct consequence of stacking fault energy.

Specimens:

- G – 6061-T6 Aluminum alloy
- K – 70/30 Brass (Cu-Zn)

Magnifications:

- 50× – grain structure overview
- 100× – grain boundaries and twins
- 500× – twin band detail
- 500× – Mg_2Si precipitates in aluminum

Etchant: Keller's reagent (G): 2 mL HF + 3 mL HCl + 5 mL HNO_3 + 190 mL H_2O | Ferric chloride (K): 5g $FeCl_3$ + 50 mL HCl + 100 mL H_2O

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens G, K (phenolic mounts)
- Al-Mg-Si precipitation / Cu-Zn stacking-fault reference
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – grain boundaries and twins → 500× – twin band detail → 500× – Mg_2Si precipitates in aluminum

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Equiaxed aluminum grains (6061-T6, Specimen G) (100×)
- ☐ Fine dark Mg_2Si precipitate dots (500×)
- ☐ Very prominent annealing twins in brass (Specimen K, almost every grain) (40–100×)
- ☐ Multiple parallel twin bands per grain (brass) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______×   Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Compare twin density in Specimen K (brass) vs. Specimen D (304 SS, Module 4). Both are FCC. Which has more prominent twins and why? (Hint: stacking fault energy.)

Answer:

Q2. The 6061-T6 designation means solution heat treated and artificially aged. What does aging do to the microstructure, and why does it increase hardness?

Answer:

Q3. At 500× on the aluminum specimen, describe any fine particles observed. What are they likely to be, and how do they strengthen the alloy?

Answer:

Q4. Using the Al-Mg-Si phase diagram concept, explain why solution treatment must precede aging. What happens if the solution treatment step is skipped?

Answer:

Step 4 — Engineering Equations & Concept Check

Lever Rule (general two-phase form)

$$W_\alpha = (C_\beta - C_0) / (C_\beta - C_\alpha)$$

- W_α — weight fraction of phase α at a given temperature
- C_0 — overall alloy composition
- C_α, C_β — compositions of phases α and β at the ends of the tie line

Critical-Thinking Questions

C1. The lever rule you used for steel in Module 3 is a special case of this general two-phase equation. What has to be true about a phase diagram for the lever rule to apply at all?

Answer:

C2. Aluminum alloys strengthen by precipitation hardening, and brass strengthens by solid-solution strengthening. Neither mechanism is described by the lever rule alone — what is the lever rule actually telling you (and not telling you) about a two-phase microstructure?

Answer:

Mechanical tests produce numbers; microstructure explains why. Hardness testing leaves a permanent indent whose geometry encodes the hardness value. Grain size is a quantitative microstructural measurement (ASTM E112) that directly correlates to strength and toughness through the Hall-Petch relationship. The Vickers-indent measurement technique you use in Q1 isn't a one-off — you'll apply it to every new alloy through the rest of the semester, building a running hardness comparison across the whole kit.

Specimens:

- A – 1018 steel (normalized) — grain size measurement
- B – 1045 steel (normalized) — grain size comparison, Vickers indent applied for Q1

Magnifications:

- 100× – grain intercept counting (ASTM E112)
- 100× – Vickers indent diagonal measurement
- 100× – grain structure comparison A vs B

Etchant: Nital 2%

Equipment & Supplies

- EXAMET-5-R-600-HD with calibrated eyepiece graticule
- Specimens A, B (phenolic mounts)
- Stage micrometer (to calibrate graticule)
- Calculator, pencil, ruler for test-line intercept counting

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – grain intercept counting (ASTM E112) → 100× – Vickers indent diagonal measurement → 100× – grain structure comparison A vs B

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Grain boundaries countable at 100× (intercept method) (100×)
- ☐ Square Vickers indent with both diagonals measurable (100×)
- ☐ Plastic zone around indent (disturbed grain boundaries) (100×)
- ☐ Grain size noticeably different between A and B (100×)

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______×   Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Measure both diagonals d_1 and d_2 of the Vickers indent placed on Specimen B. Calculate HV = 1.854 × F / d_{avg}^2 (F = 1 kgf = 9.81 N; d in mm). Report your HV result.

Answer:

Q2. Using the ASTM E112 lineal intercept method on Specimen A at 100× (field width ≈ 2.3 mm): draw a test line, count grain boundary intersections, and estimate ASTM Grain Size Number G.

Answer:

Q3. A tensile test on 1045 normalized steel gives UTS ≈ 750 MPa. Using the empirical relation UTS ≈ 3.3 × HV, does your measured HV predict a consistent UTS?

Answer:

Q4. Impact toughness (Charpy) is sensitive to grain size. If the grain size of Specimen B were refined by heat treatment, would toughness increase or decrease — and why?

Answer:

Step 4 — Engineering Equations & Concept Check

Vickers Hardness & Hardness–Strength Correlation

$$HV = 1.854 \times F / d^2 \qquad UTS\ (MPa) \approx 3.45 \times HV \quad \text{(empirical, steels)}$$

- F — applied load, kgf; d — mean diagonal length of the indent, mm
- UTS — ultimate tensile strength; HV — Vickers hardness number

Critical-Thinking Questions

C1. The Vickers equation puts diagonal length d in the denominator, squared. If two operators measure the same indent but one measures d 10% too large, will their calculated HV be too high or too low — and by roughly what factor?

Answer:

C2. The UTS ≈ 3.45 × HV correlation is empirical, not derived from first principles, and it's only reliable for steels. Why might this same multiplier give a wrong answer if applied to an aluminum alloy or a brass?

Answer:

NDT methods detect discontinuities indirectly — through magnetic flux leakage, ultrasonic reflection, or surface-penetrant bleed-out. Metallography is the destructive confirmation method NDT is designed to avoid — a distinction that matters most for parts too expensive or too critical to sacrifice. This module bridges the two approaches.

Specimens:

- A – 1018 steel (baseline reference)
- L – 360 leaded brass (discrete second-phase particles)
- P – Ti-6Al-4V titanium (aerospace-grade, NDT-first material)

Magnifications:

- 50× – full-section scan for gross defects
- 100× – discrete particle identification (L)
- 500× – particle/inclusion morphology detail

Etchant: Nital 2% (A) | Ferric chloride (L) | Kroll's reagent (P)

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens A, L, P (phenolic mounts)
- NDT method comparison reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full-section scan for gross defects → 100× – discrete particle identification (L) → 500× – particle/inclusion morphology detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Surface or near-surface void/pore located (Specimen A) (40–100×)
- ☐ Elongated MnS inclusion aligned with rolling direction (Specimen A) (100×)
- ☐ Discrete lead particles identified in Specimen L, distinguished from a true defect (100×)
- ☐ General surface condition of Specimen P noted (standard polish, no destructive prep) (50×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. In Specimen L, the lead particles are an intentional alloying addition, not a defect — yet under the scope they look like discrete dark inclusions, similar to how a real defect might appear. Which NDT method — PT, MT, or UT — would best distinguish an engineered particle from an unwanted inclusion in a real part, and why can't NDT alone always make that distinction?

Answer:

Q2. An elongated dark feature aligned with the rolling direction is visible at 100× in Specimen A. Would MT produce a strong or weak indication on this feature in a ferromagnetic part? Explain.

Answer:

Q3. Titanium aerospace components like Specimen P's alloy are almost never sectioned for metallography in service — they're inspected by NDT throughout their service life instead. Given what titanium costs and where it's used, explain why destructive metallography is reserved for failure investigations and process qualification, not routine inspection.

Answer:

Q4. Metallography destroys the part. Give two situations where destructive metallographic examination is justified over NDT in production or maintenance.

Answer:

Step 4 — Engineering Equations & Concept Check

Ultrasonic Wavelength & Detection Limit

$$\lambda = v / f \qquad d_{min} \approx \lambda / 2$$

- λ — ultrasonic wavelength in the material; v — sound velocity in the material
- f — transducer frequency; d_{min} — smallest defect size reliably detectable

Critical-Thinking Questions

C1. Using $\lambda = v/f$, if an inspector needs to detect a smaller defect, should they choose a higher-frequency or lower-frequency transducer? What trade-off in penetration depth comes with that choice?

Answer:

C2. NDT methods like UT and magnetic-particle inspection are indirect — they infer a defect exists rather than showing it directly the way metallography does. What kinds of internal features could two specimens share in a UT signature but actually be microstructurally very different under the microscope?

Answer:

Quality Standards & Defect Recognition

Quality standards require documented, reproducible measurements. ASTM E112 for grain size and ASTM E45 for inclusion rating both require systematic microscopic examination with written records — this module practices that discipline.

Specimens:

– A – 1018 steel — formal ASTM E112 grain size report

Magnifications:

– 100× – all grain-size measurements

– 100× – inclusion survey (E45)

Etchant: Nital 2%

Equipment & Supplies

- EXAMET-5-R-600-HD with calibrated graticule
- Specimen A (phenolic mount)
- ASTM E45 inclusion chart comparators (textbook or standard)
- Examination record form (write in this portfolio)
- Calculator, pencil

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – all grain-size measurements → 100× – inclusion survey (E45)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

☐ Grain boundaries consistently resolvable (three test lines) (100×)

☐ Type A inclusions (sulfide stringers) identified (100×)

☐ Type B, C, or D inclusions noted (if present) (100×)

☐ Inclusion severity rated Thin or Heavy vs. chart (100×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Complete the grain size measurement using ASTM E112 intercept method (three test lines, averaged). Record your result as ASTM Grain Size Number G.

Answer:

Q2. Perform a simplified ASTM E45 inclusion survey at 100×. For each type (A, B, C, D), record Thin or Heavy severity. Present findings in a table.

Answer:

Q3. A material certificate states 'ASTM Grain Size 7 minimum.' Does your specimen meet this requirement? (G7 ≈ 32 µm grain diameter.)

Answer:

Q4. A quality inspector must document a metallographic examination per written procedure. List four data items that must appear in a conforming examination record.

Answer:

Step 4 — Engineering Equations & Concept Check

ASTM E112 Grain Size Number

$$N = 2^{(n-1)}$$

- N — number of grains per square inch at 100× magnification
- n — ASTM grain size number

Critical-Thinking Questions

C1. Using $N = 2^{(n-1)}$, does a higher ASTM grain size number (n) correspond to larger or smaller grains? Reconcile your answer with the Hall-Petch relationship from Module 1 — does a higher n mean a stronger or weaker material?

Answer:

C2. ASTM E112 requires a documented, systematic counting procedure rather than an "eyeball estimate" of grain size. Why does reproducibility matter more in a quality-standards context than it might in a research setting where you're just trying to understand a general trend?

Answer:

 Wrought Aluminum Alloy Systems

Not every aluminum alloy is strengthened the same way. 6061 is an Al-Mg-Si (6xxx) alloy, precipitation-hardened by fine Mg_2Si particles; 2024 is an Al-Cu (2xxx) alloy, precipitation-hardened by a different phase with coarser, more visible constituent particles. Comparing both wrought alloys side by side shows how two grades can share a strengthening mechanism — aging — while producing very different microstructures and very different corrosion behavior. This week also restarts the hardness-measurement thread from Module 6 — you'll read and log HV for both G and J, the first entries in a running dataset you'll keep building all semester.

Specimens:

- G – 6061 aluminum alloy
- J – 2024 aluminum alloy

Magnifications:

- 50× – grain structure overview
- 100× – constituent particle survey
- 500× – fine precipitate and constituent particle detail
- 500× – Vickers indent diagonal measurement (G, J)

Etchant: Keller's reagent: 2 mL HF + 3 mL HCl + 5 mL HNO_3 + 190 mL H_2O

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens G, J (phenolic mounts)
- Stage micrometer or calibrated graticule (for constituent particle size estimate)
- Lens tissue, gloves
- Vickers indent pre-applied to both mounts (same technique as Module 6) — no additional equipment needed

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – constituent particle survey → 500× – fine precipitate and constituent particle detail → 500× – Vickers indent diagonal measurement (G, J)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Equiaxed grain structure visible in both specimens (50×)
- ☐ Constituent particles identified in Specimen G (6061) (500×)
- ☐ Constituent particles identified in Specimen J (2024) (500×)
- ☐ Relative particle size/density compared between G and J (500×)
- ☐ Grain boundary and twin structure noted in both (100×)
- ☐ Square Vickers indent with both diagonals measurable (G and J) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. At 500×, compare the size and number of visible constituent particles between Specimens G and J. Which alloy shows coarser, more numerous particles, and does that match what you'd expect from Al-Cu vs. Al-Mg-Si intermetallic chemistry?

Answer:

Q2. Both alloys are strengthened by aging, yet 2024 is notably less corrosion-resistant than 6061 in service (it's often sold clad with pure aluminum for exactly this reason). Based on the alloying elements each contains, propose a reason why.

Answer:

Q3. If you swapped the aging treatment of these two alloys — 2024's schedule onto 6061's composition and vice versa — would you expect similar strengthening in both? Why or why not?

Answer:

Q4. 6061 is a general-purpose structural alloy; 2024 is an aerospace skin alloy. Based on the microstructures you observed, suggest one property difference — beyond corrosion resistance — that would drive an engineer to choose one over the other.

Answer:

Q5. Using the technique from Module 6, measure the Vickers indent diagonals on Specimens G and J and calculate HV for each ($HV = 1.854 \times F / d_{avg}^2$). Record both values in your Hardness Log. Which alloy reads harder, and does that match the constituent-particle density you compared in Q1?

Answer:

Step 4 — Engineering Equations & Concept Check

Precipitation (Orowan) Strengthening

$$\Delta\sigma \approx Gb / L$$

- $\Delta\sigma$ — strength increase from precipitates
- G — shear modulus; b — Burgers vector magnitude (a lattice property)
- L — average spacing between precipitate particles

Critical-Thinking Questions

C1. Both Hall-Petch (Module 1) and the Orowan equation above describe dislocations being blocked by obstacles — grain boundaries in one case, precipitates in the other. What do d (grain diameter) and L (precipitate spacing) have in common in how they show up in each equation?

Answer:

C2. 6061 and 2024 both strengthen by precipitation hardening but reach different peak strengths. Based on the Orowan equation, what two microstructural features (visible under your microscope) would you expect to differ between a well-aged alloy and a slightly under-aged one?

Answer:

Aluminum Alloys for Forming Applications

Alloy selection for a forming process is driven as much by how an alloy responds to deformation as by its final strength. 6063 is formulated specifically for extrudability and surface finish — the standard choice for architectural window and door sections. 5052 is a non-heat-treatable Al-Mg alloy prized for its formability in sheet stampings and its corrosion resistance in marine environments. Comparing the two shows how composition is chosen around a manufacturing process, not just a target strength. You'll also take HV readings on H and I, adding two more points to the hardness dataset you started in Modules 6 and 9.

Specimens:

- H – 6063 aluminum alloy
- I – 5052 aluminum alloy

Magnifications:

- 50× – grain structure overview
- 100× – grain boundary and second-phase comparison
- 500× – fine microstructural detail
- 500× – Vickers indent diagonal measurement (H, I)

Etchant: Keller's reagent

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens H, I (phenolic mounts)
- Ruler for field-width measurement
- Lens tissue, gloves
- Vickers indent pre-applied to both mounts (same technique as Module 6) — no additional equipment needed

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – grain boundary and second-phase comparison → 500× – fine microstructural detail → 500× – Vickers indent diagonal measurement (H, I)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Equiaxed grain structure visible in both specimens (100×)
- ☐ Fine Mg_2Si precipitates identified in Specimen H (6063) (500×)
- ☐ Absence of precipitates confirmed in Specimen I (5052, solid-solution only) (500×)
- ☐ Relative grain size compared between H and I (100×)
- ☐ Square Vickers indent with both diagonals measurable (H and I) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. 6063 is heat-treatable (Mg_2Si precipitates); 5052 is not (Mg stays in solid solution). At 500×, describe what you can and cannot see in each specimen that reflects this difference.

Answer:

Q2. 6063 is formulated specifically for extrusion — lower alloy content than 6061 for smoother die flow and a better surface finish. Based on your observations, what about its microstructure would make it easier to extrude than a more heavily alloyed grade?

Answer:

Q3. 5052 work-hardens rather than age-hardens. If a sheet of 5052 were cold-stamped and then examined at 100×, what microstructural change would you expect to see compared to the as-received specimen?

Answer:

Q4. A designer needs a formed aluminum bracket for a marine application. Based on the strengthening mechanism of each alloy, which would you recommend — 6063 or 5052 — and why?

Answer:

Q5. Measure and calculate HV for Specimens H and I using the Module 6 technique, and record both in your Hardness Log. 6063 and 5052 reach their strength through different mechanisms (precipitation vs. solid-solution/work hardening) — does your HV data suggest they end up closer in hardness to each other, or closer to the 6061/2024 values you logged in Module 9?

Answer:

Step 4 — Engineering Equations & Concept Check

Hollomon Strain-Hardening Equation

$$\sigma = K \cdot \varepsilon^{n}$$

- σ — true stress; K — strength coefficient
- ε — true plastic strain; n — strain-hardening exponent (a formability indicator)

Critical-Thinking Questions

C1. A higher strain-hardening exponent n means a metal keeps getting stronger as it's deformed, rather than strength leveling off quickly. Why would a sheet-metal stamping application like 5052 favor a higher n value over a lower one?

Answer:

C2. 6063 is chosen for extrudability, not for a high n value. What does that tell you about the fact that "formable" isn't one single property — different forming processes (extrusion vs. stamping) demand different things from the alloy?

Answer:

MODULE 11 · WEEK 12 | **Machining Processes**

Machining removes material by chip formation, but the cutting zone generates heat and stress that can alter near-surface microstructure. Surface finish and subsurface integrity are quality-critical: a ground surface that looks perfect to the eye may carry thermal damage detectable only under the microscope. This week also puts your Hardness Log to work: the HV values you've already recorded for A, B, and G turn out to predict a lot about how each one machines.

Specimens:

- A – 1018 steel (baseline edge condition)
- B – 1045 steel (machined edge)
- G – 6061 aluminum (machined edge comparison)

Magnifications:

- 50× – edge overview
- 500× – near-surface grain deformation
- 500× – surface features / preparation scratches

Etchant: Nital 2% (A, B) | Keller's reagent (G)

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens A, B, G
- Surface roughness comparator card (optional reference)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – edge overview → 500× – near-surface grain deformation → 500× – surface features / preparation scratches

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Prepared edge visible and smooth at 50× (50×)
- ☐ Near-surface grain distortion or deformation layer (steel) (500×)
- ☐ Preparation scratches visible at 500× (500×)
- ☐ Comparison of near-surface condition: steel vs. aluminum (Specimen G) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. At 500×, examine the prepared edge of Specimen A. Does the near-surface grain structure differ from the interior? Describe any deformation layer you observe.

Answer:

Q2. In machining, tool wear increases cutting forces and heat. What microstructural change could this cause in a high-carbon steel workpiece — and how would it appear at 500×?

Answer:

Q3. Surface finish affects fatigue life. Identify one feature visible at 500× on any specimen that would act as a fatigue crack initiation site on a machined surface in service.

Answer:

Q4. Compare the near-surface condition of steel and aluminum specimens. Which shows more preparation-induced deformation? Why might aluminum be more susceptible?

Answer:

Q5. Pull your recorded HV values for Specimens A (1018), B (1045), and G (6061) from your Hardness Log. Rank them softest to hardest, and explain how that ranking predicts which would need the lowest cutting speed and which would show the most tool wear in machining.

Answer:

Step 4 — Engineering Equations & Concept Check

Taylor Tool-Life Equation

$$V \cdot T^n = C$$

- V — cutting speed; T — tool life (time to failure)
- n — Taylor exponent (tool-material dependent); C — constant (workpiece/tool-material dependent)

Critical-Thinking Questions

C1. Using $V \cdot T^n = C$, if cutting speed V is increased, does tool life T go up or down? Now connect this to your Hardness Log: would you expect a harder specimen (higher HV) to need a lower or higher cutting speed to hit the same tool life?

Answer:

C2. The Taylor equation only accounts for cutting speed — it says nothing about microstructure directly. Based on what you saw in Modules 9–10, what microstructural feature (not captured by V or T) could still cause two specimens with the same HV to machine very differently?

Answer:

MODULE 12 · WEEK 13 | **Copper Alloys: Strengthening Mechanisms**

Powder metallurgy and additive manufacturing are covered in this week's lecture material; this module's hands-on specimen instead continues the strengthening-mechanism thread using pure and alloyed copper. Specimen N is commercially pure copper — no alloying, no precipitates, just grain boundaries and annealing twins. Specimen O is chromium copper, a precipitation-hardening alloy that trades a little electrical conductivity for a real strength increase — the same aging mechanism you've now seen in steel (17-4PH) and aluminum (2024), applied to a third base metal entirely. You'll also log HV readings for N and O, extending the semester's hardness dataset into a third base metal.

Specimens:

- N – 110 copper (commercially pure)
- O – 182 chromium copper (precipitation-hardening)

Magnifications:

- 50× – grain structure overview
- 100× – twin band comparison
- 500× – precipitate detail (O)
- 500× – Vickers indent diagonal measurement (N, O)

Etchant: Ferric chloride: 5 g $FeCl_3$ + 50 mL HCl + 100 mL H_2O

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens N, O (phenolic mounts)
- Eyepiece graticule with grid (for twin/precipitate density comparison)
- Cross-family precipitation hardening summary card (17-4PH, 2024, chromium copper)
- Lens tissue, gloves
- Vickers indent pre-applied to both mounts (same technique as Module 6) — no additional equipment needed

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – twin band comparison → 500× – precipitate detail (O) → 500× – Vickers indent diagonal measurement (N, O)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Equiaxed grain structure visible, Specimen N (pure copper) (100×)
- ☐ Annealing twin bands visible in Specimen N (100×)
- ☐ Fine precipitate distribution identified in Specimen O (chromium copper) (500×)
- ☐ Grain/twin structure compared between N and O (100×)
- ☐ Square Vickers indent with both diagonals measurable (N and O) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. At 100×, compare grain and twin structure between Specimens N and O. Which looks 'simpler,' and why would you expect that from a single-element metal versus an alloyed one?

Answer:

Q2. You've now seen precipitation hardening in three different base metals this semester: 17-4PH (steel), 2024 (aluminum), and Specimen O (copper). List the strengthening phase in each, and state what all three approaches have in common at the mechanism level.

Answer:

Q3. Pure copper (Specimen N) has excellent electrical conductivity; the chromium addition in Specimen O reduces that conductivity somewhat. Why does adding any alloying element — even in small amounts — generally reduce electrical conductivity in a metal?

Answer:

Q4. Powder metallurgy (this week's lecture topic) sinters metal powder into a part without ever fully melting it. Based on the wrought, fully-dense specimens you've examined all semester, what microstructural feature would you expect to see in a PM part that you have not seen in any specimen this semester?

Answer:

Q5. Measure and calculate HV for Specimens N and O, and record both in your Hardness Log. Given the precipitation-hardening thread from Q2, how large is the jump from N to O — and does its size look more like the jump you'd expect from steel (17-4PH) or from aluminum (2024)?

Answer:

Step 4 — Engineering Equations & Concept Check

Matthiessen's Rule (Electrical Resistivity)

$$\rho_{total} = \rho_{thermal} + \rho_{impurity}$$

- ρ_{total} — total electrical resistivity
- $\rho_{thermal}$ — resistivity from thermal lattice vibration (temperature-dependent)
- $\rho_{impurity}$ — resistivity from solute atoms, precipitates, and defects (composition-dependent)

Critical-Thinking Questions

C1. Pure copper (Specimen N) has no alloying additions, while chromium copper (Specimen O) does. Using Matthiessen's rule, explain why O has higher electrical resistivity than N even at the same test temperature.

Answer:

C2. The same precipitates that raise chromium copper's strength (via the Orowan mechanism from Module 9) also raise its resistivity (via Matthiessen's rule here). Why can't you get the strength benefit of precipitation hardening in copper "for free," with no trade-off?

Answer:

 Brass Alloys: Composition and Microstructure

Welding process fundamentals are covered in this week's lecture material; this module's hands-on specimen instead examines how zinc content changes brass microstructure across three grades. Specimens K (cartridge brass, 70/30 Cu-Zn, revisit from Module 5) and L (free-machining brass, revisit from Module 7) return for a direct side-by-side comparison, joined by Specimen M (naval brass), seen here for the first time. K is single-phase alpha; L adds a small lead addition for machinability; M crosses into a two-phase alpha-beta structure at higher zinc content. Together they show a single compositional variable — zinc — driving a real change in phase count, not just grain size. None of K, L, or M has a logged hardness value yet, so this week you'll take the first HV reading for all three and add them to your running dataset.

Specimens:

– K – 260 brass (cartridge, single-phase alpha) (revisit, Module 5)
– L – 360 brass (leaded, free-machining) (revisit, Module 7)
– M – 464 brass (naval, duplex alpha-beta)

Magnifications:

– 100× – single-phase grain structure and twins (K)
– 100× – lead particle distribution (L)
– 100× – alpha/beta phase identification (M)
– 500× – phase boundary detail (M)
– 500× – Vickers indent diagonal measurement (K, L, M)

Etchant: Ferric chloride: 5 g $FeCl_3$ + 50 mL HCl + 100 mL H_2O

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens K, L, M (phenolic mounts)
- Cu-Zn (brass) phase diagram reference card
- Lens tissue, gloves
- Vickers indent pre-applied to all three mounts (same technique as Module 6) — no additional equipment needed

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – single-phase grain structure and twins (K) → 100× – lead particle distribution (L) → 100× – alpha/beta phase identification (M) → 500× – phase boundary detail (M) → 500× – Vickers indent diagonal measurement (K, L, M)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

☐ Equiaxed single-phase alpha grains, Specimen K (260) (100×)
☐ Annealing twin bands visible in Specimen K (100×)
☐ Discrete lead particles identified, Specimen L (360) (100×)
☐ Two distinct phases (alpha + beta) identified, Specimen M (464) (100×)
☐ Alpha/beta phase boundary morphology noted, Specimen M (500×)

□ Square Vickers indent with both diagonals measurable (K, L, and M) (500×)

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______×   Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. At 100×, identify annealing twins in Specimen K. Are they still visible in Specimens L and M, or does the added lead/beta phase make them harder to see?

Answer:

Q2. Locate the lead particles in Specimen L. Lead is essentially insoluble in the copper-zinc matrix. Explain how that insolubility is exactly what makes it useful for machinability.

Answer:

Q3. Specimen M is the only one of the three showing two distinct phases (alpha and beta) rather than one. Based on the zinc contents of these three brasses, what does that tell you about where the alpha/beta phase boundary falls on the Cu-Zn phase diagram?

Answer:

Q4. A component needs to be both machined economically and resistant to dezincification in seawater. Based on what you observed in Specimens K, L, and M, would you specify 360 or a duplex naval brass — and what's the microstructural tradeoff?

Answer:

Q5. Measure and calculate HV for Specimens K, L, and M, and record all three in your Hardness Log — their first entries. Does the duplex alpha-beta structure of M read harder than the single-phase alpha of K and L, consistent with what phase count alone would predict?

Answer:

Step 4 — Engineering Equations & Concept Check

Applying the Lever Rule to Cu–Zn Brass

$$W_\beta = (C_0 - C_\alpha) / (C_\beta - C_\alpha)$$

- W_β — weight fraction of beta phase
- C_0 — overall zinc content of the brass
- C_α, C_β — zinc content at the α and β phase-boundary compositions (from the Cu-Zn phase diagram)

Critical-Thinking Questions

C1. K (cartridge brass, ~30% Zn) is single-phase alpha, while M (naval brass, higher Zn) is two-phase alpha-beta. Using the lever-rule logic above, what has to be true about M's zinc content relative to the alpha-phase solubility limit for a second phase to appear at all?

Answer:

C2. L (free-machining brass) gets its machinability from a small lead addition, not from a phase change. Does lead show up anywhere in the Cu-Zn lever-rule calculation above? What does that tell you about the limits of a two-component phase diagram when a real alloy has three or more elements?

Answer:

 Precipitation Hardening: A Cross-Family Comparison

Weld metallurgy and failure analysis are covered in this week's lecture material. This module instead revisits three specimens examined earlier in the semester — Specimen F (17-4PH stainless steel, Module 4), Specimen J (2024 aluminum, Module 9), and Specimen O (182 chromium copper, Module 12) — to directly compare how the same strengthening mechanism, precipitation hardening, plays out across three different base metals, crystal structures, and precipitate chemistries. You'll take your first HV reading on F here, then pull your already-logged values for J and O from your Hardness Log — turning three separate weekly measurements into a single cross-family strength comparison.

Specimens:

- F – 17-4PH stainless steel (revisit, Module 4)
- J – 2024 aluminum (revisit, Module 9)
- O – 182 chromium copper (revisit, Module 12)

Magnifications:

- 500× – precipitate / constituent particle comparison across all three
- 500× – Vickers indent diagonal measurement (F)

Etchant: As previously used — Kalling's No. 2 (F), Keller's reagent (J), ferric chloride (O). No re-etch required if specimens are still in prepared condition.

Equipment & Supplies

- EXAMET-5-R-600-HD
- Specimens F, J, O (revisit — retrieve from kit, phenolic mounts)
- Calibrated graticule (precipitate size comparison)
- Precipitation hardening cross-family summary card
- Lens tissue, gloves
- Vickers indent pre-applied to Specimen F's mount (same technique as Module 6) — no additional equipment needed

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 500× – precipitate / constituent particle comparison across all three → 500× – Vickers indent diagonal measurement (F)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

- ☐ Precipitate morphology re-examined in Specimen F (17-4PH) (500×)
- ☐ Precipitate morphology re-examined in Specimen J (2024) (500×)
- ☐ Precipitate morphology re-examined in Specimen O (chromium copper) (500×)
- ☐ Relative precipitate size/density ranked across all three (500×)
- ☐ Base crystal structure noted for each (martensitic vs. FCC) (100×)
- ☐ Square Vickers indent with both diagonals measurable, Specimen F (500×)

Step 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Fill in a short table from memory and your earlier observation notes: for each of F, J, and O, name the base metal, the precipitate phase (if known), and whether it appeared fine/hard-to-resolve or coarse/clearly visible at 500×.

Answer:

Q2. All three of these alloys go through the same basic aging heat-treatment sequence conceptually: solutionize, quench, age. Explain in your own words what each of those three steps accomplishes, using one of the three specimens as your example.

Answer:

Q3. Precipitation hardening increases strength without changing the base crystal structure (martensitic in F, FCC in J and O). Why is a fine, closely-spaced precipitate more effective at blocking dislocation motion than a coarse, widely-spaced one?

Answer:

Q4. If all three specimens were left at room temperature for years ('natural aging') instead of artificially aged, would you expect them to reach the same final strength as the artificially aged condition? Explain why or why not, using what you know about diffusion rates at room temperature.

Answer:

Q5. Measure and calculate HV for Specimen F — its first entry in your Hardness Log. Then pull your already-logged values for J and O. Using rough annealed baselines (304 stainless ≈ 150 HV, 6061 ≈ 30 HV, pure copper ≈ 40 HV), rank the three families by relative strength increase from precipitation hardening. Which family gained the most?

Answer:

Step 4 — Engineering Equations & Concept Check

Revisiting Orowan Strengthening Across Three Base Metals

$$\Delta\sigma \approx Gb / L \quad \text{(Module 9)}$$

- G — shear modulus (different for each base metal — steel, aluminum, copper)
- b — Burgers vector; L — interparticle spacing

Critical-Thinking Questions

C1. F (17-4PH steel), J (2024 Al), and O (182 Cu) all strengthen by the same Orowan mechanism, but steel's shear modulus G is roughly 3× aluminum's and copper's. Does that alone guarantee 17-4PH will show the largest $\Delta\sigma$ of the three? What else would you need to know about L to answer with confidence?

Answer:

C2. You now have logged HV values for F, J, and O in your Hardness Log. Do the relative HV rankings match what the Orowan equation alone would predict from G, or do differences in precipitate spacing (L) between the three alloys appear to dominate instead?

Answer:

MODULE 15 · WEEK 16 | Course Capstone & Synthesis

Corrosion mechanisms and protection methods are covered in this week's lecture material and the course reference pages — including a direct callback to Specimens D and E (304 vs. 316) from Module 4, since molybdenum content and PREN are exactly why 316 outperforms 304 in chloride environments. This final module is a synthesis exercise: no new specimen, just your own observations from across the semester, organized into one final comparison. Your Hardness Log — built from Modules 6, 9, 10, 12, 13, and 14 — also comes back into play as evidence for the Unit 3 link in this week's synthesis question.

Specimens:

– No new specimen — synthesis exercise using observation notes and sketches from Modules 1–14

Magnifications:

– N/A — reference your own notes and sketches from prior modules

Etchant: N/A

Equipment & Supplies

- Portfolio notes, sketches, and answered questions from Modules 1–14
- PREN / corrosion reference card (for the Module 4 callback)
- Precipitation hardening summary from Module 14
- Pencil, calculator
- This portfolio, for final review

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Gather your observation tables, sketches/photographs, and answered questions from Modules 1–14.
2. Re-read your notes on Specimens D and E (304 vs. 316, Module 4) before answering the PREN-related question.
3. Re-read your notes on Specimens F, J, and O (Module 4, 9, 12) before answering the precipitation-hardening synthesis question.
4. No microscope session is required for this module — it is a written synthesis exercise.
5. Review the full course arc (Units 1–5) before writing your synthesis answer.

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

Step 1 — Observation Checklist

☐ Crystal structure identified for at least one BCC, FCC, and HCP specimen (from notes)

☐ Twin presence/absence compared across at least three specimens (from notes)

☐ Strengthening mechanism identified for at least three specimens (from notes)

☐ Precipitation-hardening thread traced across F, J, O (from notes)

☐ Five-unit synthesis chain outlined with specimen citations

Step 2 — Sketch or Photograph

Specimen: ________________   Magnification: ______×   Etchant: ________________

(attach photograph or draw the field of view at the magnification that best shows the key microstructural feature)

Step 3 — Short-Answer Questions

Q1. Across every specimen you examined this semester, identify one example each of a BCC structure, an FCC structure, and an HCP structure (Specimen P, titanium, Module 7). List the specimen and family for each.

Answer:

Q2. You observed annealing twins in several FCC specimens (304, 316, brass) but never in the BCC carbon steels. Explain why, in terms of stacking fault energy and crystal structure, twinning is common in some metals and essentially absent in others.

Answer:

Q3. You also encountered three strengthening mechanisms this semester: solid-solution strengthening (5052, pure copper — no precipitates), precipitation/age hardening (17-4PH, 2024, chromium copper), and grain-boundary strengthening via Hall-Petch (present to some degree in every specimen). For any one specimen, identify which mechanism(s) applied and explain your reasoning.

Answer:

Q4. Connect the course's five units using your own microscopy observations as evidence: How does atomic structure/bonding (Unit 1) → alloy family and composition (Unit 2) → mechanical testing and inspection (Unit 3) → manufacturing process and alloy selection (Unit 4) → service behavior including corrosion resistance (Unit 5) form a single chain? Cite at least three specimens by letter as evidence.

Answer:

Q5. Compile your full Hardness Log (A, B, G, H, I, J, K, L, M, N, O, F) into one ranked list from softest to hardest. Pick any two adjacent alloys from different families and explain, using composition and microstructure, why their hardness values ended up so close despite being different metals.

Answer:

Step 4 — Engineering Equations & Concept Check

Semester Equation Toolkit (recap)

Hall-Petch: $\sigma y = \sigma 0 + ky/\sqrt{d}$ | PREN = %Cr + 3.3(%Mo) + 16(%N) | $HV = 1.854F/d^2$, UTS ≈ 3.45×HV | Orowan: $\Delta\sigma \approx Gb/L$

– Definitions as introduced in Modules 1, 4, 6, and 9 — see those pages if you need a refresher.

Critical-Thinking Questions

C1. Hall-Petch and Orowan strengthening both block dislocations, but PREN doesn't describe strength at all — it describes corrosion resistance. Using your Module 4 specimens (D vs. E, 304 vs. 316), explain why a metal can be re-engineered to resist corrosion without changing its strengthening mechanism at all.

Answer:

C2. Across all 15 modules, which single equation from the toolkit above connects most directly to what you can actually see under the EXAMET-5 microscope, and which one is hardest to connect to a visual observation? Defend your answer using at least two specimens from your semester.

Answer:

Appendix A — Class Specimen Kit & Field Reference

The kit contains 16 pre-prepared, labeled metallographic specimens. Do not regrind or re-etch. Return all specimens in the labeled foam tray.

Magnification Field Widths — EXAMET-5-R-600-HD

- **50× (≈ 8.0 mm):** Full cross-section overview, gross specimen scan
- **100× (≈ 2.3 mm):** Grain counting (ASTM E112), inclusion survey (ASTM E45), indent identification, grain aspect ratio, ferrite/pearlite and phase fraction comparison
- **500× (≈ 0.46 mm):** Precipitates and constituent particles, twin boundary detail, phase boundary detail (duplex brass, precipitation-hardening grades), pearlite lamellar spacing

Specimen Summary

- **A —** 1018 steel (normalized) · Mount: Hot-compression phenolic · Etchant: Nital 2%
- **B —** 1045 steel (normalized) · Mount: Hot-compression phenolic · Etchant: Nital 2%
- **C —** 1075 steel (normalized) · Mount: Hot-compression phenolic · Etchant: Nital 2%
- **D —** 304 austenitic stainless steel · Mount: Hot-compression phenolic · Etchant: Marble's reagent
- **E —** 316 austenitic stainless steel · Mount: Hot-compression phenolic · Etchant: Marble's reagent
- **F —** 17-4PH precipitation-hardening stainless · Mount: Hot-compression phenolic · Etchant: Kalling's No. 2
- **G —** 6061 aluminum alloy · Mount: Hot-compression phenolic · Etchant: Keller's reagent
- **H —** 6063 aluminum alloy · Mount: Hot-compression phenolic · Etchant: Keller's reagent
- **I —** 5052 aluminum alloy · Mount: Hot-compression phenolic · Etchant: Keller's reagent
- **J —** 2024 aluminum alloy · Mount: Hot-compression phenolic · Etchant: Keller's reagent
- **K —** 260 brass (cartridge, 70/30) · Mount: Hot-compression phenolic · Etchant: Ferric chloride
- **L —** 360 brass (leaded, free-machining) · Mount: Hot-compression phenolic · Etchant: Ferric chloride
- **M —** 464 brass (naval, duplex alpha-beta) · Mount: Hot-compression phenolic · Etchant: Ferric chloride
- **N —** 110 copper (commercially pure) · Mount: Hot-compression phenolic · Etchant: Ferric chloride
- **O —** 182 chromium copper (precipitation-hardening) · Mount: Hot-compression phenolic · Etchant: Ferric chloride
- **P —** Titanium Grade 5 (Ti-6Al-4V) · Mount: Hot-compression phenolic · Etchant: Kroll's reagent

Semester Hardness Log

- **A —** 1018 steel (normalized) · First recorded: Module 6 (Wk 7) · HV: ___________
- **B —** 1045 steel (normalized) · First recorded: Module 6 (Wk 7) · HV: ___________
- **C —** 1075 steel (normalized) · First recorded: — · HV: —
- **D —** 304 austenitic stainless steel · First recorded: — · HV: —
- **E —** 316 austenitic stainless steel · First recorded: — · HV: —
- **F —** 17-4PH precipitation-hardening stainless · First recorded: Module 14 (Wk 15) · HV: ___________
- **G —** 6061 aluminum alloy · First recorded: Module 9 (Wk 10) · HV: ___________
- **H —** 6063 aluminum alloy · First recorded: Module 10 (Wk 11) · HV: ___________
- **I —** 5052 aluminum alloy · First recorded: Module 10 (Wk 11) · HV: ___________
- **J —** 2024 aluminum alloy · First recorded: Module 9 (Wk 10) · HV: ___________

- **K —** 260 brass (cartridge, 70/30) · First recorded: Module 13 (Wk 14) · HV: ___________
- **L —** 360 brass (leaded, free-machining) · First recorded: Module 13 (Wk 14) · HV: ___________
- **M —** 464 brass (naval, duplex alpha-beta) · First recorded: Module 13 (Wk 14) · HV: ___________
- **N —** 110 copper (commercially pure) · First recorded: Module 12 (Wk 13) · HV: ___________
- **O —** 182 chromium copper (precipitation-hardening) · First recorded: Module 12 (Wk 13) · HV: ___________
- **P —** Titanium Grade 5 (Ti-6Al-4V) · First recorded: — · HV: —

Grading Summary

Each module is graded on five components: Observation table completeness, sketch/photograph quality, short-answer accuracy, critical-thinking reasoning, and kit return/handling. The 15 weekly modules together are worth 75 points; the holistic end-of-semester portfolio review is worth 25 points — totaling 100 points for the semester.

- **Observation table** — 1 pt — Credit for each feature located and checked
- **Sketch / photograph** — 1 pt — Labeled with specimen, magnification, etchant
- **Short-answer questions (Step 3, 4–5)** — 1 pt — Credit apportioned across all questions in the set
- **Critical-thinking questions (Step 4, 2)** — 1 pt — Equation applied correctly and reasoning tied to the observed microstructure
- **Kit return & handling** — 1 pt — On time, no damage, handled by edges
- **Total per module** — 5 pts
- **15 modules total** — 75 pts — 5 pts × 15 modules
- **End-of-semester portfolio review** — 25 pts — Holistic; all 15 modules as a complete document
- **TOTAL** — 100 pts